Carson R. Malpass

Linkedin.com/in/carsonmalpass
Carsonrmalpass@gmail.com
Littleton, CO 80123
(720) 537-7474

Mechanical Engineering at the University of Colorado Boulder
Incoming Product Development FCG Engineer at Ford Motor Company

EDUCATION

Bachelor of Science in Mechanical Engineering, University of Colorado Boulder *August 2022 - May 2026*
- GPA: 3.35 | Minor in Engineering Management | Fundamentals of Engineering - Mechanical

PROFESSIONAL EXPERIENCE

Student Apprentice – *University of Colorado, Boulder* *August 2024 - Present*
- Provide leadership for 250-person MCEN 2000 course facilitating breakout groups, resume reviews, and critiquing work.
- Support industry events and contribute to new career initiatives and curriculum development.

Mechanical Engineering Intern – *Manifest Technologies, Boulder* *April 2024 – October 2024*
- Co-designed the Athena printer with focus on the 80/20 frame structure, LED latent-curing system, wiring, and assembly.
- Developed and tested proprietary High-Definition Volumetric Additive Manufacturing technology (HD VAM).
- Operated HD VAM 3D printers to analyze and enhance image sets, part characteristics, and dimensional accuracy.
- Designed optical image systems to obtain laser beam measurement data, along with necessary fixturing for integration.
- Built and aligned various z-fold laser systems involving DMD's, irises, tip/tilt mirrors, and magnification lenses.
- Worked on establishing post-processing procedures involving handling, acid-cleaning, light-curing, and documentation.
- Constructed numerous SOPs for tests and printing with the HD VAM system.

Special Projects Technician Intern – *Malin Space Science Systems, San Diego* *May 2023 - August 2023*
- Focused on Mars Stationary Orbiter (MSO) camera development through vacuum chamber test design and construction.
- Performed microscopic reflow soldering and dimensional tolerance inspections on flight-certified PCB hardware.
- Gained hands-on experience in ESD protection, cleanroom practices, and specialized handling of space-qualified hardware.
- Repaired and operated resin and FDM 3D printers to produce ESD-safe components for spaceflight vacuum testing.

Mechanical Engineering Intern – *M3Robotics Lab, Colorado School of Mines* *May 2021 - August 2021*
- Helped create an acoustic classification method for determining mining tool wear using frequency spectra.
- Designed circuitry mounting and enclosures for an automated underground drill oriented to promote miner safety.
- Repaired FDM 3D-printers and optimized machine settings to comply with product specifications.

LEADERSHIP EXPERIENCE

President, Project Manager, Chief Chassis Engineer – CU Racing Team, *Boulder* *August 2022 - Present*
- Oversee $150,000+ budget and 300-member team spanning Endurance Road Racing, Formula SAE IC, Formula SAE EV.
- Support the creation of the team's first ever Formula SAE EV platform and assist in acquiring a new facility.
- Responsible for a 150-member team of business, chassis, powertrain, electronic, and aerodynamic teams as PM.
- Led recruitment, interviews, industry relations, and managed team socials reaching 128,000 monthly views.
- Applied SolidWorks for CAD, FEA, renderings, and PDM management; delivered technical component presentations.
- Managed 50-member chassis team for the 2024 competition year. Facilitated design, integration, and assembly.
- Designed and fabricated steering wheel and column, control arms, seat, floor, and firewall for multiple vehicles.
- Represented the team at the 2023-2025 Formula SAE competitions achieving 20th, 63rd, and 31st place finishes.

Manager, Engineer – *Warren Tech Shell Eco-Marathon Race Team, Lakewood* *August 2020 - May 2022*
- Managed engineering peers and build of hydrogen-powered competition vehicle.
- Competed at Indianapolis Motor Speedway, leading the program to passing technical inspection for the first time.
- Designed and assembled brake system, Lexan body, egress system, livery, and hydrogen fuel cell integration.
- Led team socials and coordinated competition deliverables and events.

Design Finalist – *NASA Hunch National Competition, Denver* *August 2020 - May 2022*
- Used astronaut-driven requirements to design a hands-free food bite helmet dispenser for use during spacewalks.
- Developed lunar supply pod shoes for astronauts using Nomex and custom features per NASA specifications.
- Underwent a series of technical presentations and product reviews to boards of NASA personnel and professionals.

SKILLS

- **Part Manufacturing** – 3D-Printing, Mills, Lathes, Woodworking, Metalworking, MIG/TIG Welding, Sand Casting
- **CAD** – SolidWorks (CSWA), SW FEA, SW Visualize, Engineering and Manufacturing Drawings, AutoCAD, Revit, Sketchup
- **Software & Electronics** – EES, MATLAB, C++, Soldering, PCB Manufacturing, Arduino, Eagle, LabVIEW
- **Business** – Project and Operations Management, Marketing and Recruitment, Client and Industry Relations